

OBE Course Outline: MAT-211

Part A- Introduction

1. Course No./ Course Code: MAT-211
2. Course Title: Numerical Analysis
3. Course Type (Core Course/Electives/...): Core Course
4. Level/Term and Section: 2/1 (2nd year 1st Semester)
5. Academic Session: Spring-2023
6. Course Instructor: Rina Paul
7. Prerequisite (If any): Differential and Integral Calculus (MAT 111), Linear Algebra and Differential equation (MAT 112)
8. Credit Value: 3
9. Contact Hours: 3 Hours/Week
10. Total Marks: 100
11. Course Objectives and Course Summary

Course Objectives –

1. To overview numerical solutions and its necessity. Description of various errors involve in a solution and how to compute them.
2. Introducing finite difference operators. Illustrate the concept of interpolation and methods of interpolation.
3. Derive several numerical methods for solving an algebraic and transcendental equation. Introducing the algorithm of the methods
4. To facilitate the numerical methods for solving a system of linear equations.
5. Introducing various methods for numerical differentiation and integration. Overview their algorithms.
6. Introducing Numerical methods for solving ordinary differential equations (ODE). Solution of IVP's and BVP's.

Course Summary-

The main purpose of the course is to introduce the fundamentals of numerical analysis that are mainly used in engineering. It deals with the theory and application of numerical approximation techniques as well as their computer implementation. It covers computer arithmetic, solution of nonlinear equations, interpolation, and approximation, numerical integration and differentiation, solution of differential equations, and matrix computation.

12. Course Learning Outcomes (CLO)

By the end of this course, the students should be able to –

CLO 1	Differentiate between analytic and numerical solutions and their uses. Compute absolute, relative, percentage errors and rate of convergences.
CLO 2	Concept on forward, backward, central, shift and average operator. Ability to approximate a function by interpolation technique by using Newton's and Lagrange's interpolation formula.
CLO 3	Apply numerical methods (such as: Bisection, Iteration, False position and Newton-Raphson) to solve algebraic and transcendental equations. Create a computer program to use a particular method.
CLO 4	Apply Tri diagonal, LU Decomposition, Jacobi and Gauss-Seidel's method for solving a system of linear equation. Create computer code to use these methods.
CLO 5	Apply Newton's Forward and Backward polynomials for numerical differentiation. Apply Trapezoidal, Simpson's 1/3, Simpson's 3/8 th rule for Numerical Integration. Create computer programs for these methods.
CLO 6	Use of Euler's method, Modified Euler's method, Runge-Kutta Method for solving ordinary differential equations (ODE). Generate computer programs for using these methods.

13. Program Learning Outcomes (PLO)

After graduation, students will get the fundamental concepts of computer science as well as software engineering. They will be able to identify computational problems and devise algorithmic solutions. They will be able to build software that can be employed to solve real-world problems with economic, environmental, social, political, health and safety, manufacturability and sustainability constraints. Besides, they will achieve communication and managerial skills. The nine significant Program Learning Outcomes (PLO) of the BSc. in Software Engineering Programss are mentioned point wise.

- PLO1: Students will be able to demonstrate broad knowledge of historical events, periods along with their significance; act according to their professional and ethical responsibility. Moreover, students will get good command in Bangla as well as in academic English.
- PLO2: Students will be able to identify computational problems and devise algorithmic solutions

- PLO3: Students will be able to apply critical reasoning to issues through independent thought and informed judgment and make decisions using an evidence-based approach. Students will be able to identify problems appropriate for research, make a critical analysis of the literature, analyze research data, and draw logical conclusions.
- PLO4: Students will achieve computational, algorithmic and technological problem-solving fluency.
- PLO5: Students will be able to design, develop, document and test software using modern techniques.
- PLO6: Students will gain the aptitude to communicate and collaborate effectively with multidisciplinary terms in large-scale and complex projects.
- PLO7: Students will be able to utilize modern computing languages of diversified domains, platforms and tools necessary for engineering practice.
- PLO8: Students will be able to engage in lifelong learning and self-development.
- PLO9: Students will be able to learn professional and organizational aptitude which will make them competitive as well as collaborative for employment.

14. Mapping/Alignment of CLOs with Program Learning Outcomes (PLO) (Optional)

	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9
CLO 1		✓		✓					
CLO 2		✓		✓			✓		
CLO 3		✓		✓			✓		
CLO 4		✓		✓			✓		
CLO 5		✓		✓			✓		
CLO 6		✓		✓			✓		

Part B- Content of the Course

15. Course Content (Topics to be covered)

Topics	Time Frame	Specific Outcome(s)	Teaching Strategies and Suggested Activities	Teaching Aids/Tools/ Material	Assessment Technique	Align ment with CLOs
What is Numerical analysis? Why this branch is important to study for Computer science students. Errors in numerical calculations. Error: Definitions, sources, examples	Week 1	- To be aware of the reasons to study Numerical Analysis - To learn about errors in numerical calculation.	Real life example. Interactive discussion. Demonstration	Reference Book, Power point Presentation.		CLO 1
Error Analysis- Propagation of Error. A general error formula. Taylor series and reminders.	Week 2	- To learn how error propagation occurs - Deriving formulas of error propagation - Calculating Error using Taylor's series	Demonstration, Problem solving in pairs.	Reference Book, Hand note.		CLO 1
Root finding: The bisection method and the iteration method.	Week 3	- To learn how to determine roots of Various functions numerically (Algebraic, Transcendental, logarithmic etc.) using Bisection method and Iteration method. - To write computer codes using the methods.	Demonstration, Lab work, Interactive discussion.	Programming Computers. Power point slides, Reference book.	Assignment - 1	CLO 3
Root Finding - The method of false position. Newton-raphson method. Roots of polynomials.	Week 4	- To learn how to determine roots of Various functions numerically (Algebraic, Transcendental, logarithmic etc.) using discussed methods. - To write computer codes using the methods.	Demonstration, Lab work, Interactive discussion.	Programming Computers. Power point slides, Reference book.		CLO 3

Methods of approximation theory: Polynomial interpolation: Equal interval: Newton's forward difference method, Backward Difference method, and Central Difference method.	Week 5	- To interpolate a polynomial using given coordinates of equal intervals - learn to use forward difference and backward difference - Ability to write computer codes to use the methods -	Demonstration, Lab work, Interactive discussion.	Programming Computers. Power point slides, Reference book.	Group Presentation	CLO 2
Unequal interval: Lagrange form, divided formula for interpolation.	Week 6	- To interpolate a polynomial using given coordinates of unequal intervals - learn to use Lagrange interpolation and divided difference formula - Ability to write computer codes to use the methods	Demonstration, Lab work, Interactive discussion.	Programming Computers. Power point slides, Reference book.	Class Test – 1 (on the lessons of Week 1 –Week 5)	CLO 2
	Week 7				MID-TERM EXAMINATION	
Sol. of system of linear equation: Gaussian elimination, pivoting strategy, Gauss Jordan elimination, Iterative Methods.	Week 8	- Ability to solve linear equations - Understanding pivoting strategy to apply in Gaussian Elimination method. - Learn Gauss Jordan method and understand the comparison between the two methods	Demonstration, Problem solving in pairs.	Reference book, Handout		CLO 4
Solution of tridiagonal systems, LU decomposition, Matrix inverse, Eigen value problems for matrices.	Week 8	- Learn to decompose a matrix tri-diagonally. - Learn to determine inverse of matrix - Learn to determine eigenvalues of a matrix	Demonstration, Problem solving in pairs.	Reference book, Handout		CLO 4
Numerical solution of ODE – Euler's method, Improved Euler's Method, Modified Euler's Method.	Week 9	- Ability to solve an Ordinary differential equation using Euler's method, modified Euler's method and improved Euler's method - Ability to write computer codes to use the methods	Demonstration, Problem solving.	Reference book, Handout		CLO 6
Numerical solution of ODE – Runge Kutta method of higher order.	Week 10	- Learn to use higher order Runge-Kutta method to solve higher order - Ability to write computer codes to use the methods	Demonstration, Problem solving. Lab work	Programming Computers. Power point slides, Reference book.	Assignment - 2	CLO 6
Numerical Differentiation: Newton's forward difference method, Backward difference method.	Week 11	- Learn to calculate value of higher order derivatives at a point for any function numerically. - Ability to write computer codes to use the methods	Demonstration, Problem solving.	Reference book, Handout	Group Presentation	CLO 5
Numerical Integration: Trapezoidal method. Simpson's method. Weddle's method;	Week 12	- Learn to evaluate integrals numerically. - Ability to write programs to evaluate integrals numerically.	Demonstration, Problem solving. Lab work	Programming Computers. Power point slides, Reference book.		CLO 5
Review Class-1	Week 13				Class Test - 02	
Review Class - 2	Week 14				Lab Test	

Part C- Assessment and Evaluation

16. Assessment Strategy (Quiz, Mid Term, Assignment, Project, Report, Class Test, Spot Test, Presentation, etc. and Semester Final):

Class Test:

There will be two class tests of 10 marks each in the semester. One will be taken for the mid-term examination and the other for the final examination.

Average marks of the two tests will be added with the final marks. If any student misses the exam for any emergency (justification required), s/he will be able to sit for the make-up exam. But, once the make-up exam is held, there will be no opportunity to sit for this test again.

Assignment and presentation:

There will be two assignments to submit in each semester. Two group presentations will be held too, one in each semester. Two of the assignments should be completed individually. No late submission of the assignments will be accepted.

Mid-Term, Make-up, Final and Supplementary Exam:

Mid Term examination will be held on the 7th or 8th week of the semester. There is an opportunity to sit for a make-up exam if the examinee cannot attend the mid-term examination due to an emergency. Student must justify the reason for his/her absence. If any student fails to secure 40% marks in the 60 marks evaluation before the final (Attendance: 10 + Class Test: 10 + Assignment/Presentation/Quiz/Oral: 20), he/she would not be able to sit for the final exam and will have to Retake the course. However, if any student fails to pass the 40 marks final exam (written), he/she will have an opportunity to appear in a supplementary exam within a month to pass the course.

17. Evaluation Policy (Grading System)

Grades will be calculated as per the university grading structure and individual student will be evaluated based on the following criteria with respective weights.

(i) Attendance	10%
(ii) Tutorials	10%
(iii) Assignment/Presentation/Oral	20%
(iv) Mid-Term Examination (1 hour 30 minutes)	20%
(v) Final Examination (2 hour)	40%
Total	100%

18. Attendance Evaluation –

Percentage	Marks
95%-100%	10
90%-94%	9
85%-89%	8
80%-84%	7
75%-79%	6
70%-74%	5
65%-69%	4
60%-64%	3
55%-59%	2
50%-54%	1
<50%	0

18. Make-up Procedures

There is an opportunity to sit for a make-up exam if the examinee cannot attend the mid-term examination due to an emergency. Student must justify the reason for his/her absence. There is no such opportunity in the final examination. The make-up exam will be held within few days after the mid-term examination, and the course teacher will provide a different question for this exam.

Part D-Learning Resources

1. Text Books

Introductory Methods of Numerical Analysis by S.S Sastry

Numerical Analysis for Scientists and Engineers: Theory and C Programs by Madhumangal Pal

2. Reference Books

Numerical Methods for Engineers

Raymond P. Canale and Steven C. Chapra

